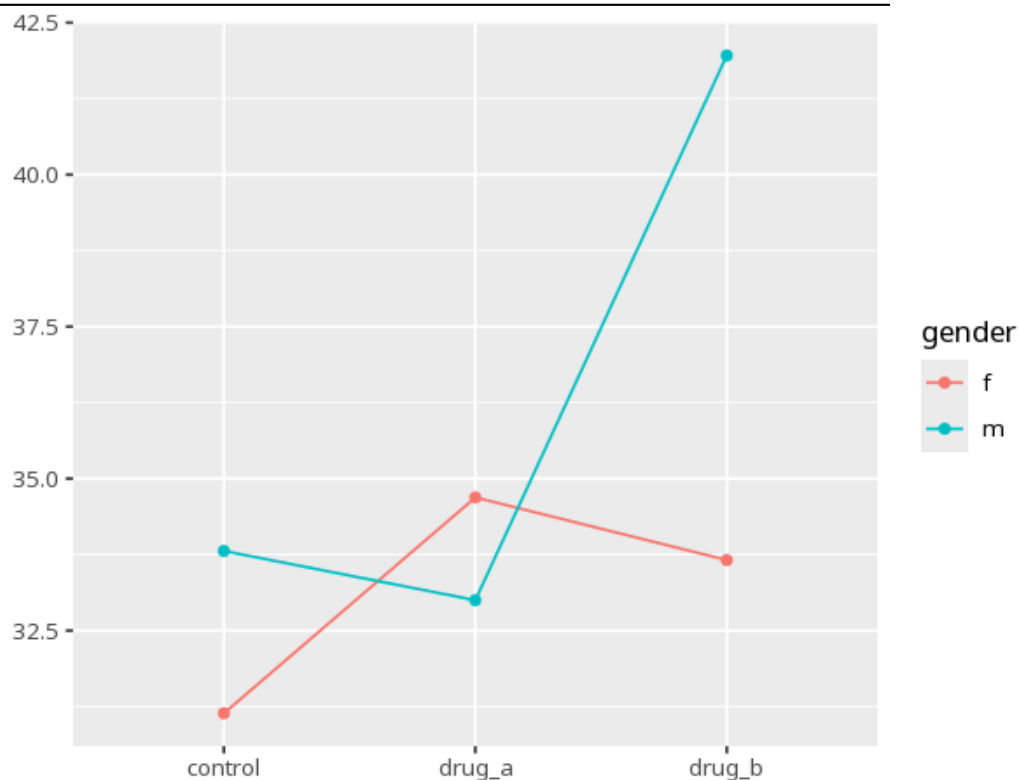


1 Factorial ANOVA

Factor A $\times$ B	N	Mean	Std. Dev.
control $\times$ f	10	31.14	8.22
drug_a $\times$ f	10	34.69	7.22
drug_b $\times$ f	10	33.66	8.27
control $\times$ m	10	33.81	6.79
drug_a $\times$ m	10	33.00	6.86
drug_b $\times$ m	10	41.96	4.65

Source	SS	df	MS	F	p	partial η^2 [95% CI]
group	307.07	2	153.53	3.04	.056	0.10 [0.00, 1.00]
gender	143.53	1	143.53	2.84	.098	0.05 [0.00, 1.00]
group $\times$ gender	250.84	2	125.42	2.48	.093	0.08 [0.00, 1.00]



Each factor's F/p is its main effect; the A $\times$ B row tests whether the effect of one factor depends on the other. A significant interaction usually takes priority — read it from the interaction plot before the main effects. A significant effect for any factor with 3+ levels tells you the levels differ somewhere, not which ones — use the post-hoc / simple-effects (emmeans) table to find the specific pairs. Your significance threshold (α) is 0.05.

A two-way ANOVA gave A×B interaction $F(2,54)=2.48$, $p = .093$, partial $\eta^2=.08$ [.00, 1.00].

Statistical basis: Factorial ANOVA (main effects + interaction) (Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver & Boyd.); Estimated marginal means for interaction / simple-effects follow-up (Lenth RV (2024). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package.); η^2 effect-size benchmarks (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.).